Experiment 2

Upload and analyze a CSV dataset and perform data preprocessing and visualization.

Code 1

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

file_path = 'sales_data.csv'
df = pd.read_csv(file_path)

print(df.head())
print(df.isnull().sum())

df['Sales'].fillna(df['Sales'].mean(), inplace=True)
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)

print(df.describe())

product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()

print(product_summary)

plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()

df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)
sales_over_time = df.groupby('Date').agg({'Sales': 'sum'}).reset_index()

plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.show()

pivot_table = df.pivot_table(
    values='Sales',
    index='Region',
    columns='Product',
    aggfunc=np.sum,
    fill_value=0
)
print(pivot_table)

correlation_matrix = df.corr(numeric_only=True)
print(correlation_matrix)

plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

Output

```
         Date    Product  Sales  Quantity Region
0  01-01-2023  Product A    200         4  North
1  02-01-2023  Product B    150         3  South
2  03-01-2023  Product A    220         5  North
3  04-01-2023  Product C    300         6   East
4  05-01-2023  Product B    180         4   West
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64
            Sales   Quantity
count   16.000000  16.000000
mean   237.500000   5.375000
```

```
std     64.031242   1.746425
min    150.000000   3.000000
25%    187.500000   4.000000
50%    225.000000   5.500000
75%    302.500000   7.000000
max    340.000000   8.000000
     Product  Sales  Quantity
0  Product A   1350        33
1  Product B    850        17
2  Product C   1600        36
```

Output

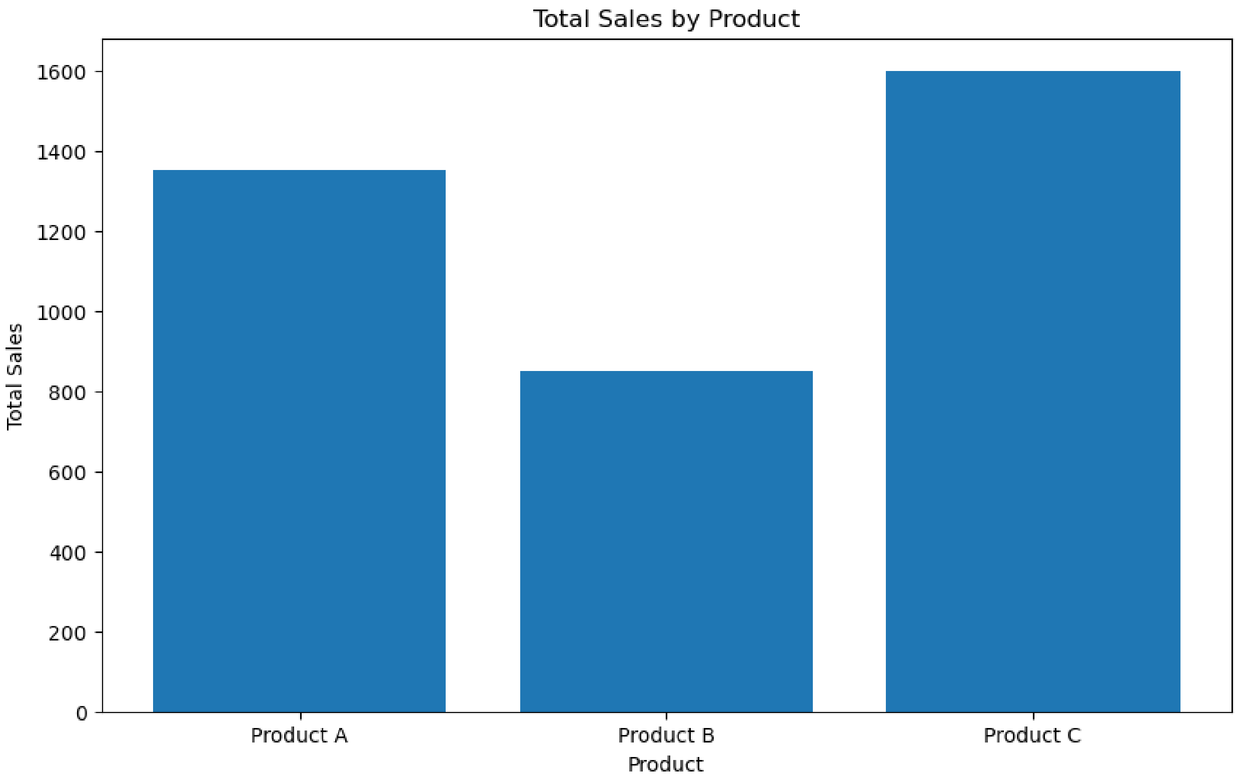
```
C:\Users\vsros\AppData\Local\Temp\ipykernel_16952\3353328291.py:12: FutureWarning: A value is trying to be set on a cop
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which wo

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df


  df['Sales'].fillna(df['Sales'].mean(), inplace=True)
```

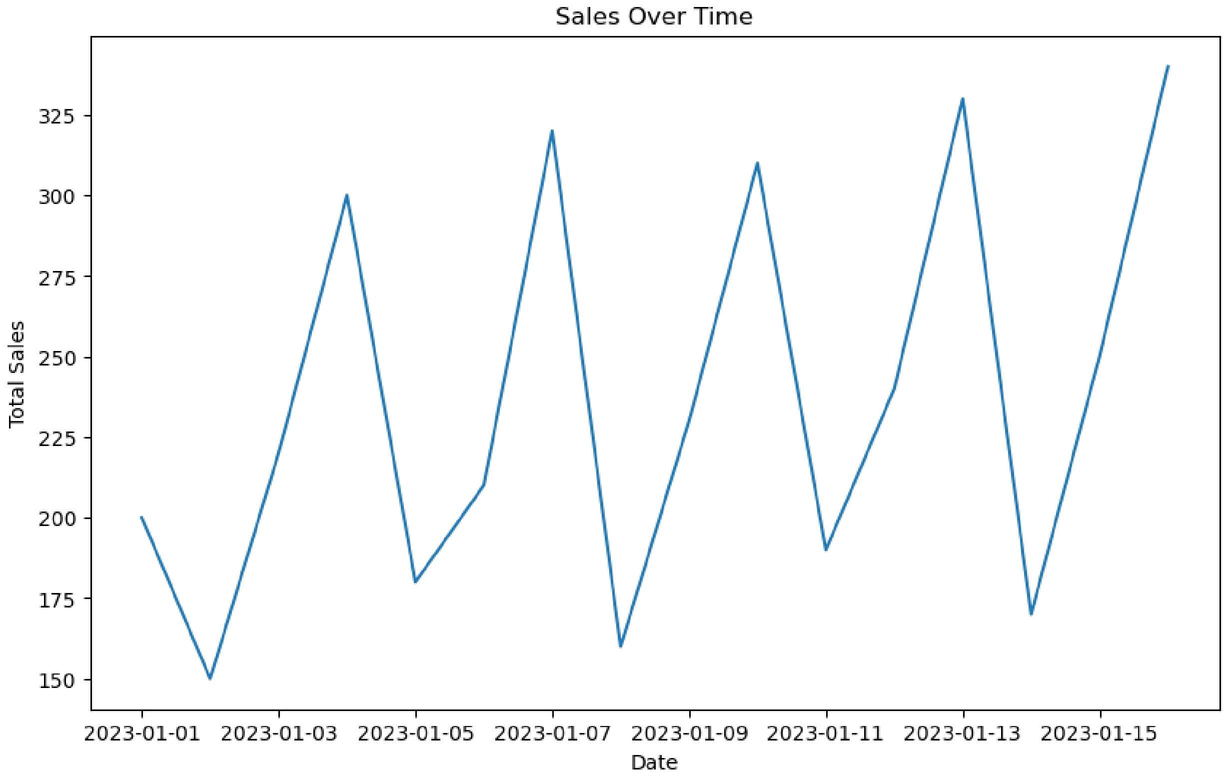
Output

```
<Figure size 1000x600 with 1 Axes>
```



Output

```
<Figure size 1000x600 with 1 Axes>
```



Output

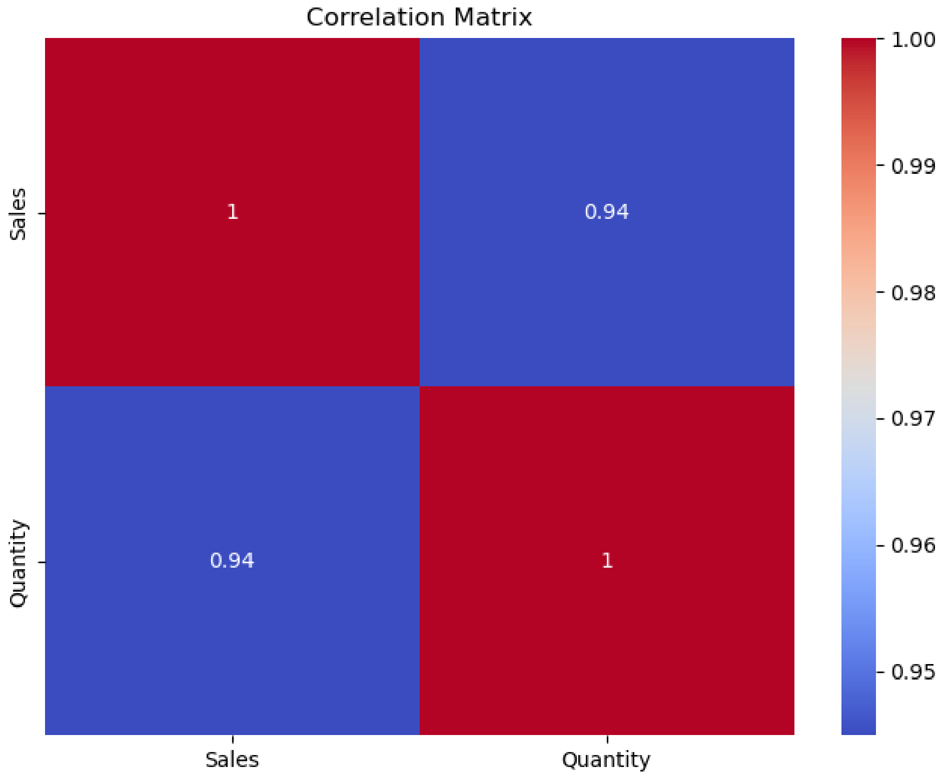
```
Product  Product A  Product B  Product C
Region
East             0          0       1600
North         1350          0          0
South            0        480          0
West             0        370          0
             Sales  Quantity
Sales     1.000000  0.944922
Quantity  0.944922  1.000000
```

Output

```
C:\Users\vsros\AppData\Local\Temp\ipykernel_16952\3353328291.py:41: FutureWarning: The provided callable <function sum
  pivot_table = df.pivot_table(
```

Output

```
<Figure size 800x600 with 2 Axes>
```


Correlation Matrix
Sales
Quantity
1
0.94
0.94
1
Sales
Quantity
1.00
0.99
0.98
0.97
0.96
0.95